

Dollar Dynamics, not Dollar Weaponization: Assessing the 2022 Russia Sanctions Effect on Onshore-Offshore RMB Pricing

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Abstract

I examine whether the February 2022 Russia sanctions shock, the largest peacetime asset freeze in history, caused a permanent structural shift in offshore renminbi (RMB) market conditions. Using daily trading data spanning January 2020 through December 2025 and AR(1)-GARCH(1,1) models with Student-t errors, I find no statistically significant sanctions impact in the CNY-CNH basis spread at either the announcement date or the implementation date.

Movement in the spread is dominated by dollar dynamics. Daily changes in the DXY dollar index are the only meaningful predictor in the model.

The conclusion is a null result with substantive content. The Federal Reserve's continued increase in interest rates, not economic demand for RMB, explains the movement in the spread. GARCH persistence indicates that pricing volatility in the spread nonetheless became structurally elevated after the shock and did not dissipate within the sample window. The sanctions shifted the uncertainty regime without moving the level.

1. Introduction

On February 26, 2022, the United States, European Union, and allied governments announced the freezing of approximately \$300 billion in Russian central bank foreign exchange reserves.

Within days, major Russian banks were disconnected from SWIFT, Russia's sovereign wealth fund was frozen, and the ruble entered freefall. The event marked an unprecedented use of dollar-denominated financial infrastructure as an economic weapon, and it was watched closely by every government that holds dollar reserves.

The economic logic of a consequent shift toward offshore RMB is straightforward. If dollar reserves can be frozen unilaterally, the cost of holding dollars rises for any government that perceives itself as a potential future target. China's offshore RMB market, centered in Hong Kong, offers the most developed alternative to dollar-denominated international finance. The question is whether the 2022 shock actually moved behavior in offshore RMB markets or if the movement was driven by the contemporaneous dollar cycle.

This paper provides direct high-frequency financial market evidence on that question. The CNY-CNH spot spread captures whether offshore yuan is trading at a premium or discount to onshore yuan, a measure of the price of offshore RMB access.

The identification challenge is that the sanctions coincided almost exactly with aggressive Federal Reserve rate hikes. The Federal Reserve raised the federal funds rate from 0.25% in March 2022 to 5.50% by July 2023, driving broad dollar changes. As such, because CNY-CNH spread is constructed from two exchange rates both quoted in yuan-per-dollar terms, dollar dynamics mechanically affect it. Any analysis without this channel will attribute dollar movement to economic demand.

The finding is that once dollar dynamics are properly controlled for, the sanctions effect disappears. Neither the announcement nor the implementation shock is statistically distinguishable from zero. The movement following the 2022 Russia sanctions was effectively a dollar story in disguise.

2. Institutional Background

The offshore RMB market: The renminbi trades in two distinct but related markets. The onshore market (CNY) operates within mainland China under capital account controls administered by the People's Bank of China (PBOC). The offshore market (CNH) operates primarily in Hong Kong and is freely tradeable by international participants. The two markets are linked through arbitrage channels including the Stock Connect programs, and PBOC bilateral currency swap lines with foreign central banks, but capital controls prevent perfect convergence.

The persistent gap between CNY and CNH exchange rates reflects the residual friction from these controls.

The 2022 shock: On Saturday February 26, 2022, the US and EU announced comprehensive sanctions including the freezing of Russian central bank assets, SWIFT disconnection of major Russian banks, and suspension of Russia's ability to transact in dollar, euro, and pound sterling. A first formal EU sanctions package entered into force on approximately March 12, 2022. Both dates fall on weekends; I identify the first available trading day following each announcement as the shock date for market analysis. Simultaneously with these events, the Federal Reserve began its most aggressive rate hiking cycle since 1980, raising the federal funds rate from 0.25% in March 2022 to 5.50% by July 2023. This creates a critical identification challenge: separating the sanctions driven component of any RMB market movement from the dollar driven component of the same movements.

3. Data

3.1 Sample

The preshock period (January 2020 through February 25, 2022) serves as the baseline. All data sourced from Bloomberg Professional terminal. Daily observations from January 2, 2020 through December 31, 2025 yield 1,455 raw observations, reducing to 1,454 after first-differencing, and 1,453 in the regression.

3.2 Dependent Variable: CNY-CNH Basis Spread

Constructed as the percentage deviation of onshore yuan from offshore yuan, expressed in basis points:

$$\text{cny_cnh_spread}_t = [(\text{CNY}_t - \text{CNH}_t) / \text{CNH}_t] \times 10,000$$

CNY is onshore RMB pricing, **CNH** is offshore RMB pricing, both denominated in USD terms. A negative spread indicates offshore CNH is trading at a premium, one dollar buys fewer yuan offshore than onshore, meaning offshore yuan is more expensive. Increased offshore RMB demand would be expected to push this spread negative.

Bloomberg tickers: CNY REGN Curncy (onshore daily fixing rate) and CNH BGN Curncy (offshore BGN composite).

Summary statistics: mean = -7.34 bps, standard deviation = 16.84 bps, range = [-116.89, 69.28] bps.

3.3 Control Variables

VIX Index (VIX Index): the CBOE Volatility Index, measuring implied equity market volatility. Controls for global risk-off episodes that could independently affect the dependent variable as investors reduce all emerging market exposures.

DXY Dollar Index (DXY Curncy): the trade-weighted US dollar index. The critical control for this analysis. The dollar's broad appreciation cycle in 2022 mechanically affects any spread quoted in yuan-per-dollar terms. Without this control, sanctions effects and dollar effects are indistinguishable in the spot FX data.

US 3-Month Treasury Yield (USGG3M Index): captures Federal Reserve policy stance. Complements DXY by separating the level of the dollar from the direction of US rates, as these can diverge during risk events.

3.4 Stationarity and Data Treatment

All variables are tested for stationarity using the Augmented Dickey-Fuller (ADF) test. However, standard ADF tests are known to fail in the presence of structural breaks, potentially failing to reject a false unit root null (Perron, 1989). Spread variables are therefore additionally tested with the Zivot-Andrews (1992) structural break test, which permits one unknown break in the sample, appropriate given the prior expectation of a February 2022 break.

Results:

Variable	ADF p-value	ZA p-value	Decision
CNY-CNH Spread	0.0002	0.0019	Stationary: model in levels
VIX	0.0002	X	Stationary: include in levels
DXY	0.4726	X	Nonstationary: first-difference
US 3M Yield	0.8515	X	Nonstationary: first-difference

Given this discovery of non-stationarity, I first-difference DXY and the US 3-month yield before inclusion, since mixing I(1) regressors with a stationary dependent variable produces spurious inference.

First differences of DXY and US3M are confirmed stationary (ADF $p < 0.0001$ for both). VIX passes ADF and is included in the model.

Holiday treatment. Chinese onshore markets observe extended closures (Chinese New Year, National Day). These represent genuine non-trading days, not missing data. I apply last-observation-carried-forward imputation with a maximum gap tolerance of two consecutive days for control variables sourced from continuously trading markets. Gaps exceeding two days are deleted. Trading days are identified using the Shanghai Stock Exchange calendar. This element was particularly challenging to work through given the unsynchronized trading days between US markets and Greater China markets.

4. Empirical Methodology

4.1 Model Specification

The model estimates AR(1)-GARCH(1,1) [Auto Regressive + Generalized Autoregressive Conditional Heteroskedasticity] via maximum likelihood with robust covariance estimation, using the ARCH framework of Engle (1982) and the GARCH extension of Bollerslev (1986).

The model follows the intervention analysis framework of Box and Tiao (1975), which formalizes the effect of a known external event on an autocorrelated time series by incorporating dummy variables representing the intervention directly into the mean equation.

The general system models time-varying volatility as a function of past squared residuals and past conditional variance where:

Mean equation:

$$y_t = \alpha + \phi \cdot y_{t-1} + \beta_1 \cdot \text{shock}_{1_t} + \beta_2 \cdot \text{shock}_{2_t} + \gamma \cdot X_t + \varepsilon_t$$

Variance equation:

$$\sigma_t^2 = \omega + a \cdot \varepsilon_{t-1}^2 + b \cdot \sigma_{t-1}^2$$

y_t is the relevant spread

y_{t-1} is the AR(1) lag

shocks are event dummies

X_t is the control vector

ε_t are innovations distributed as standardized Student-t with estimated degrees of freedom v .

Model Breakdown:

AR(1) [Auto-Regressive, 1 Lag]:

Fundamentally, AR model calculates the expected baseline level of the spread today by combining yesterday's price ($\phi \cdot y_{t-1}$), the sanctions event ($\beta_1 \cdot \text{shock}_{1_t} + \beta_2 \cdot \text{shock}_{2_t}$), and the macro controls ($\gamma \cdot X_t$).

AR(1) specifies that the model looks back exactly 1 trading day. This model states that **today's spread** (y_t) is heavily influenced by **yesterday's spread** (y_{t-1})

GARCH(1,1) [Generalized Auto-Regressive Conditional Heteroskedasticity]:

This model predicts today's risk or market volatility (σ^2_t) by establishing baseline noise (ω), yesterday's shock ($a \cdot \varepsilon^2_{t-1}$), yesterday's lingering volatility ($b \cdot \sigma^2_{t-1}$).

This is what captures volatility clustering (the idea that calm days follow calm days, and chaotic days follow chaotic days)

(1,1): This specifies the lag structure for volatility prediction.

The first 1: indicates that the model looks back at 1 lag of yesterday's unexpected surprise/shock (ε^2_{t-1})

The second 1: indicates that the model looks back at 1 lag of yesterday's total market variance (σ^2_{t-1})

The AR(1) term controls for the strong autocorrelation inherent in financial spreads.

GARCH(1,1) accounts for volatility clustering; the variance equation allows periods of elevated uncertainty to cluster rather than assuming constant residual variance.

The Student-t distribution accommodates fat tails. Standard normal distribution assumes that extreme market crashes almost never occur, but by specifying Student-t distribution, the model handles 'fat-tails' where estimates degrees of freedom parameter (ν) dictates the weight of said tails.

The estimated ν of 4.24 indicates extreme moves occur far more frequently than any normal distribution would predict.

4.2 Shock Variables

I specify **two shock dates** corresponding to the sanctions announcement and the formal implementation package.

CNY-CNH model: Step dummies. CNY-CNH is found to be stationary, therefore a typical 'step dummy' is introduced for both shock dates.

Variables equal zero before the shock date and one from the shock date onward permanently. A significant coefficient indicates the average level of the spread permanently shifted, the direct test of sustained demand change.

Both announcement dates (February 26 and March 12, 2022) fall on Saturdays. I shift both to the following Monday, February 28 and March 14 respectively, as the first trading day on which markets could respond.

5. Results

5.1 Primary Result: CNY-CNH Spot Channel: Dollar Dynamics, Not Sanctions

Table 1 presents the CNY-CNH results.

Neither shock dummy achieves statistical significance: the announcement coefficient is -1.65 bps ($p = 0.505$) and the implementation coefficient is -0.85 bps ($p = 0.735$). These results cannot be distinguished from zero.

The dominant predictor of the CNY-CNH spread is the daily change in the DXY dollar index: $\beta = -6.22$ bps per unit ($p < 0.001$). This coefficient has a straightforward interpretation. The 2022 dollar rally, driven by Federal Reserve rate increases from March 2022 to July 2023, was the primary force moving the CNY-CNH spot differential throughout the sample.

Every one-point DXY appreciation compressed the spread by 6.22 basis points. Over the approximately 18-point DXY rally from January to September 2022, this accounts for over 110 basis points of spread movement, swamping any sanctions specific component. This result is not a failure. It is a finding.

The naive precontrol result showing CNY-CNH compression after February 2022 was real in the raw data but confounded. Once the dollar cycle is properly absorbed, the sanctions component of spot RMB dynamics is indistinguishable from zero. Dollar monetary policy, not geopolitical demand for RMB, drove the 2022 spot exchange rate differential.

Table 1. CNY-CNH Basis Spread: AR(1)-GARCH(1,1) Estimates

Variable	Coefficient	Std Error	p-value	95% CI
Mean equation				
Constant	1.0471	1.332	0.432	[-1.563, 3.658]
CNY_CNH[1]	0.633	0.029	<0.001***	[0.576, 0.690]
Announcement(dummy)	-1.652	2.479	0.505	[-6.512, 3.207]
Implement (dummy)	-0.855	2.525	0.735	[-5.804, 4.095]
VIX	-0.072	0.061	0.235	[-0.191, 0.047]
Δ DXY	-6.218	0.777	<0.001***	[-7.741, -4.696]
Δ US 3M yield	-0.625	9.604	0.948	[-19.448, 18.199]
Variance equation				
ω (constant)	4.120	3.022	0.173	[-1.802, 10.042]
α (ARCH term)	0.143	0.051	0.005***	[0.044, 0.243]
β (GARCH term)	0.850	0.055	<0.001***	[0.743, 0.957]
Distribution				
ν (degrees of freedom)	4.239	0.500	<0.001***	[3.258, 5.220]

Notes: Δ denotes first differences, applied to DXY and US 3M Treasury yield after both failed ADF stationarity tests. Standard errors are robust. GARCH persistence ($\alpha + \beta$) = 0.993
N = 1,453. $R^2 = 0.378$. Adjusted $R^2 = 0.375$. Log-likelihood = -5,554.28. AIC = 11,130.6. BIC = 11,188.6.

Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

5.2 Interpreting the Findings

The CNY-CNH spread is the most visible and widely cited measure of onshore-offshore RMB divergence. It is what traders watch in real time and what journalists cite when discussing de-dollarization. Its behavior in 2022 was driven almost entirely by the Federal Reserve. This is actually important information: the widely discussed narrative that 2022 spot RMB dynamics reflected economic demand shifts was primarily a dollar story in disguise.

This does not contradict Chupilkin et al. (2023), who document substantial sanctions driven CNY invoicing in Russian trade over the same period using customs microdata. Trade invoicing and FX pricing are distinct fields and evidence from one should not be read as evidence for the other.

6. Limitations

Identification.

This paper identifies the sanctions effect using before-after variation in a single aggregate time series. It cannot rule out all contemporaneous confounders, including any PBOC intervention in offshore markets during 2022, which could have independently affected currency dynamics.

Measurement scope. The CNY-CNH spread measures the price of offshore RMB access, not the quantity of RMB adoption. The bilateral country-level SWIFT payment data used by Bahaj and Reis (2026) would measure adoption directly but is proprietary and unavailable for this analysis

Statistical power. Failure to reject the null is not evidence for the null.

Abandoned interbank specification:

An interbank funding channel test using the SHIBOR-HIBOR differential was attempted and abandoned. The HIBOR series available on Bloomberg Terminal is HKD rather than the necessary CNH, making the spread cross-currency rather than onshore-offshore. Also, because the HKD is pegged to the USD, it captures any inherent changes to the USD. The resulting spread therefore contains a US rate component by construction, meaning the US 3M yield would be mechanically related to the dependent variable. It would be possible to construct a synthetic offshore Hong Kong based RMB rate for similar analysis but this is left to future work.

7. Conclusion

This paper asks whether the 2022 Russia sanctions shock created a detectable and permanent shift in offshore RMB market conditions.

The spot FX channel, CNY-CNH, shows no sanctions effect once dollar strength is controlled for. The 2022 spot RMB dynamics were a dollar story driven by Federal Reserve tightening.

Claims that spot RMB appreciation in 2022 reflected geopolitical demand are not supported by the data once concurrent monetary policy is properly absorbed.

The volatility findings reinforce this interpretation. Near-unit-root GARCH persistence in the model indicates that pricing uncertainty in offshore RMB markets became structurally elevated after the shock and did not dissipate. The sanctions created a volatility regime change without a level shift.

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Replication code available at github.com/spm01. Contact the author for data availability. All analysis conducted in Python 3.13 using the arch, statsmodels, and pandas libraries.